Ask Small VLMs Only What They Can See: A Semantic-to-Metric Agent for Embodied Navigation

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Abstract

Multimodal LLM agents navigate by giving one model every role: it reads the image and its own history, keeps track of the room, and writes the plan. Small open VLMs, the models a robot can run on board, fail in this role. With the EB-Navigation benchmark's agent, Qwen3-VL-4B solves 55.0% of the test episodes, and 25.0% when the target starts behind it. A controlled diagnosis rules out motion planning, since a geometric planner given the target position solves 91.3% of the same episodes, and recognition, since the model boxes a visible target in 69.3–78.7% of single views. What fails is keeping spatial state in generated text. We therefore recast the VLM from planner to sensor. S2M (Semantic-to-Metric) is a training-free navigation agent that asks the VLM only what the current image can answer, whether the target is visible and where, and keeps all state in metric form. Its hypothesis memory lifts each answer with depth and pose, accumulates evidence across distinct viewpoints, and withdraws it when a hypothesis should be visible but is not detected; a search policy and an occupancy-map planner turn the memory into actions. With the same 4-bit model on one 8 GB GPU, S2M reaches 79.4% (+24.4 pp), 82.1% on Long-horizon episodes, and 76.5% from RGB alone, above both fine-tuned 4B agents and seven of ten training-free agents built on larger models. With Qwen2.5-VL-3B it lifts success from 20.3% to 70.0%. On RoboTHOR ObjectNav, which requires an explicit stop, it reaches 59.6%, above ten published zero-shot agents.

1. Introduction

Multimodal LLM (MLLM) agents follow open navigation instructions by giving one model every role: at each step it reads the image and the text history, infers where the target is, and writes the next actions [1]. This design works at frontier scale. On EB-Navigation, GPT-4o succeeds in 57.7% of the episodes, while open models with at most 8B parameters reach 14.0–29.3% overall and 0–5% on the Long-horizon subset, in which the target starts behind the agent. Yet these small models are the ones a robot can run on board or on a consumer GPU.

Existing remedies leave the model in charge. Training-free agents surround it with memory, reflection, skill libraries, scene graphs, or tools [2–6], but the model still writes the plan, and the reported gains rely on 8B to frontier-scale models. Fine-tuned 4B agents [7, 8] require training and remain below 60%. Modular object-navigation systems keep a metric map [9–11] but depend on dedicated detectors.

We first ask what breaks when a small VLM drives the agent. Motion planning does not: a geometric planner given the target position solves 91.3% of the test episodes. Recognition is imperfect but usable: Qwen3-VL-4B boxes a visible target in 69.3–78.7% of single views. The failure lies in the role the model is given. Asked for a plan at every step, it must carry where the target was and which way it faces through generated text (Fig. 1a), and nothing checks this text against the world. Prompted agents therefore lose targets they have seen, collide with furniture, and never turn toward a target behind them (Sec. 3).

A small VLM is thus a good semantic sensor and a poor state keeper, so we build the agent around the model rather than inside it. S2M (Semantic-to-Metric, Fig. 1b) is a training-free navigation agent with three parts. *Perception* asks the VLM two questions per frame, whether the target is visible and where it is in the image, and spends a magnified crop or a tiled view only when the answer is weak. *Memory* lifts each answer with depth and pose into a metric target hypothesis, accumulates evidence only from distinct viewpoints, and withdraws it when a hypothesis should be visible but is not detected. *Action* commits to a confident hypothesis or otherwise searches in the order of an optimal-search index, and an occupancy-map planner computes every move. Each part removes one diagnosed failure: the memory keeps a seen target, the map prevents collisions, and the look-around finds a target behind the agent.

Changing the model's role changes the outcome (Fig. 2), with the profile that the diagnosis predicts. With Qwen3-VL-4B at 4-bit, S2M succeeds in 79.4% of the EB-

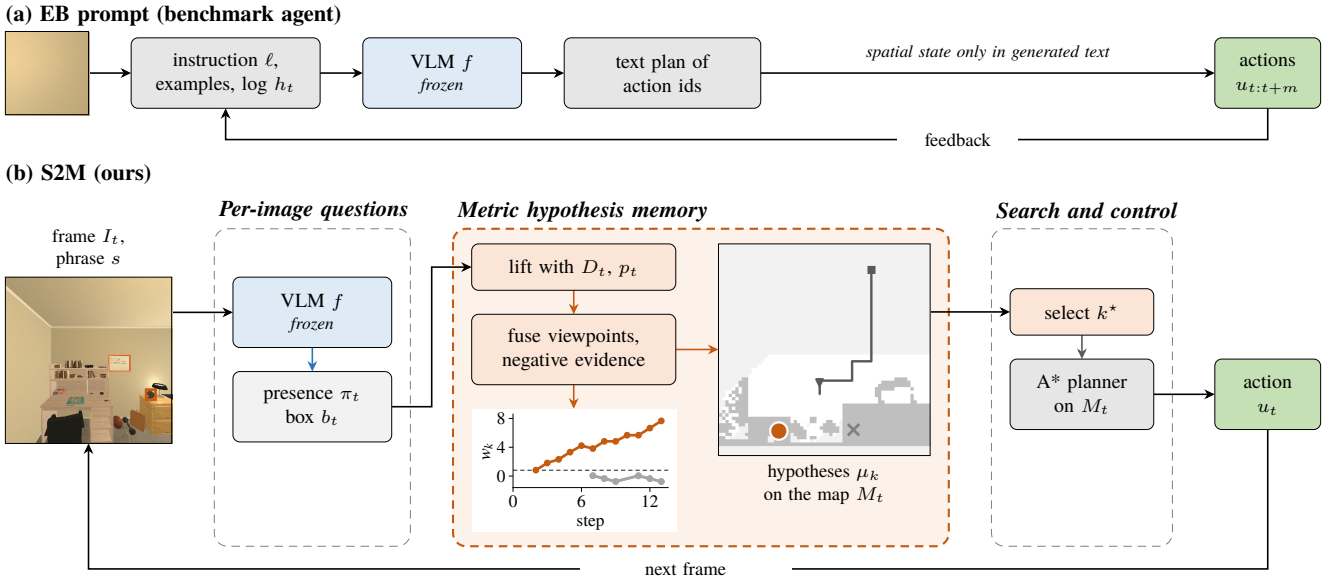


Figure 1. **(a)** The EB prompt maps I_t, the instruction ℓ, and the history h_t to a text plan $u_{t:t+m}$ with the VLM f. **(b)** S2M asks f for the presence π_t and box b_t of the phrase s, lifts them with depth D_t and pose p_t into hypotheses μ_k with evidence w_k, selects $k^\star$, and plans u_t on the map M_t. Thumbnails: a Long-horizon test episode; the target (orange) passes the commit threshold (dashed) and two look-alikes (grey) fall below zero.

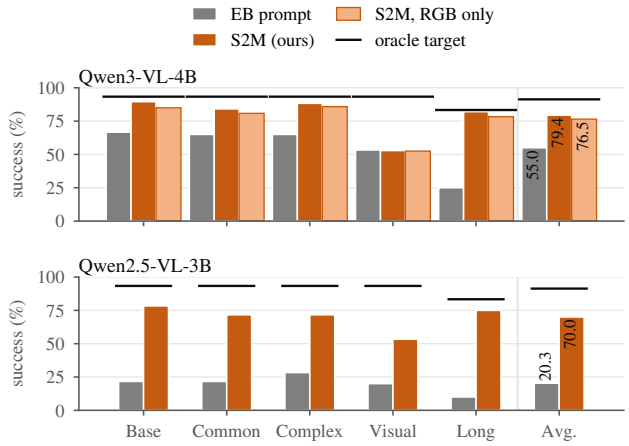


Figure 2. EB-Navigation test success with the same local model. The benchmark's agent (grey) against S2M (orange) and S2M without a depth sensor or simulator pose (light orange). Ticks: geometric planner given the true target position.

Navigation test episodes, +24.4 pp over the benchmark's agent with the same model, and in 82.1% of the Long-horizon episodes. Its average exceeds both fine-tuned 4B agents and seven of ten training-free agents built on 8B to frontier-scale models. With Qwen2.5-VL-3B [12] the gain grows to +49.7 pp, so the agent depends far less on the size of the model. From RGB alone S2M reaches 76.5%, and on RoboTHOR ObjectNav [13], which requires an explicit stop, 59.6%.

Contributions. (1) A controlled diagnosis of small-VLM navigators, showing that they fail at keeping spatial state in text rather than at recognition or motion planning, and the principle it yields: ask the model only what the current image can answer, and keep the state in metric form. (2) S2M, a training-free agent built on this principle, whose metric hypothesis memory fuses per-image answers across viewpoints with positive and negative evidence, and whose per-frame floor scale lets it run from RGB alone. (3) Evidence on the full EB-Navigation and RoboTHOR splits, with two model generations, that S2M removes the diagnosed failures and lets a 4-bit 4B model match agents built on far larger models.

2. Related Work

MLLM agents for embodied tasks. EmbodiedBench [1] evaluates multimodal LLMs on high- and low-level embodied tasks through a standardized action interface; related benchmarks probe embodied decision making and navigation reasoning [14, 15]. Agents are improved by embodied pre-training, online reinforcement learning, or policy-level reasoning [16–19], or at test time with reflection, experience, memory, or tools [2–4, 20–25]. On EB-Navigation, models of 8B and more also call mapping tools [4] or read a bird's-eye view [26]. All of these agents keep the model as the planner and the reader of its own memory. In S2M the VLM is only a sensor: its answers feed an explicit metric

estimator and planner, which is what lets a 4B model drive the agent.

Zero-shot object navigation and semantic memory. Modular zero-shot ObjectNav combines open-vocabulary perception, a map of explored space, and a planner. CoW [9] pairs open-vocabulary localization with frontier exploration; later systems build vision-language value maps [10], scene graphs reasoned over by an LLM [11, 27], goal graphs [28], world models [29], cognitive state machines [30], or structured representations queried by a VLM [31], and instruction-following agents ground language in maps and scene graphs [32–34]. 3D scene graphs, semantic maps, language fields, and embodied memories attach semantics to geometry [35–42]. Closest to our memory, ApexNav fuses target-centric detections over time [43], and AECNav treats expected but absent detections as negative evidence [44]. S2M sits at the minimal end of this spectrum: a small generative VLM is its only perception model, and its memory is a short list of metric target hypotheses rather than a dense semantic map or scene graph.

Verification and search. Spatially explicit prompting [45–47], verification, test-time computation, and calibrated uncertainty [48–53] improve VLM decisions in embodied tasks. S2M spends extra queries, a magnified crop or a tiled view, only where the evidence is weak, and chooses among candidate targets with the probability-to-cost index of optimal sequential search [54].

Geometry from images. Learned reconstruction and depth priors recover geometry from RGB [55–60], and sensors can change the conclusions of embodied evaluations [61–63]. Monocular visual odometry fixes its scale from the ground plane and the known camera height [64]; S2M applies the same constraint to every frame of a monocular metric depth network. Concurrently, Li *et al*. [65] run Qwen3-VL-4B zero-shot and accumulate its answers as per-direction scores in a bird's-eye field for vision-language navigation; S2M lifts boxes to metric target hypotheses for object navigation.

3. Problem and Diagnosis

Task. An episode gives an instruction ℓ that identifies one object with floor position $x^\star \in \mathbb{R}^2$. At step t the agent observes an RGB frame I_t, depth D_t, a pose $p_t = (x_t, \theta_t, \phi_t)$ of floor position, heading, and camera tilt, and the success flag o_{t-1} of its last action, and it chooses $u_t \in \mathcal{U}$: four 0.25 m translations, two 90° turns, and two 30° tilts. The episode is successful if

$$\exists\, t \le T : \ \|x_t - x^\star\| \le 1\,\mathrm{m}, \qquad T = 20. \tag{1}$$

EB-Navigation [1] poses this task in single AI2-THOR rooms [66]; its test set has five subsets of 60 episodes. *Base* names the category, *Common* describes the object by its use, *Complex* embeds it in a longer instruction, *Visual* describes only its appearance, and *Long* repeats Base with the agent turned away from the target.

The benchmark's agent. The benchmark's own agent, the *EB prompt*, maps the image and its text history $h_t = (u_{0:t-1}, o_{0:t-1})$ directly to a plan,

$$(u_t, \ldots, u_{t+m}) = f(I_t, \ell, h_t), \tag{2}$$

which is executed until an action fails. All spatial state, where the target was, what the room contains, and how the agent has turned, must be carried by h_t and the generated reasoning. S2M factorizes the same decision:

$$\begin{aligned} (\pi_t, b_t) &= f(I_t, s), \\ \mathcal{B}_t = g(\mathcal{B}_{t-1}, \pi_t, b_t, D_t, p_t), &\qquad u_t = \rho(\mathcal{B}_t, p_t), \end{aligned} \tag{3}$$

where s is a target phrase, f answers only questions about I_t, and the metric belief $\mathcal{B}_t$ and the planner ρ hold all state and every decision (Sec. 4 and Fig. 3). The configuration of S2M was fixed on a development split of 150 episodes that we built with the benchmark's generator on 30 iTHOR rooms not used by the test set.

Planning is not the bottleneck. Replacing the target belief of S2M by the true $x^\star$, with the same occupancy map and planner, solves 91.3% of the 300 test episodes (93.3% on each of the first four subsets and 83.3% on Long).

Single views are informative but unreliable. For each test episode of the first four subsets we render the start view and the three other 90° headings and query the VLM with the phrase s. Qwen3-VL-4B places a box with IoU$\geq$0.3 on the visible target in 78.7, 69.3, 77.3, and 49.3% of the views on Base, Common, Complex, and Visual (75 visible instances each). It also errs the other way: on views that do not show the target it still places a box in 15–19% of the cases for named categories and in 41% for appearance-only phrases. A single answer is therefore evidence, not a decision.

The prompted agent loses spatial state. We label each failed episode by its first matching outcome: *never seen*, *near miss* (within 1.5 m), *lost after seen* (out of view in the last five steps), *blocked* (30% of actions rejected), oscillation, or approach failure. Of the 135 episodes that the EB prompt fails with Qwen3-VL-4B, 43 lose a target that was in view, 26 are blocked by collisions, and 34 end within 1.5 m (Fig. 4b); in 12 Long episodes the agent never brings the target into view.

What the diagnosis predicts. If small VLMs fail because their role makes them carry spatial state through text, an agent that keeps this state outside the model, with the same model and the same actions, should (i) remove the never-seen and blocked failures and reduce the lost-after-seen ones; (ii) gain most on Long, where the target must be remembered across turns; (iii) gain least on Visual, where single-view recognition is weakest and stays the limit; and (iv) depend far less on the size of the model, which no longer plans (Sec. 5).

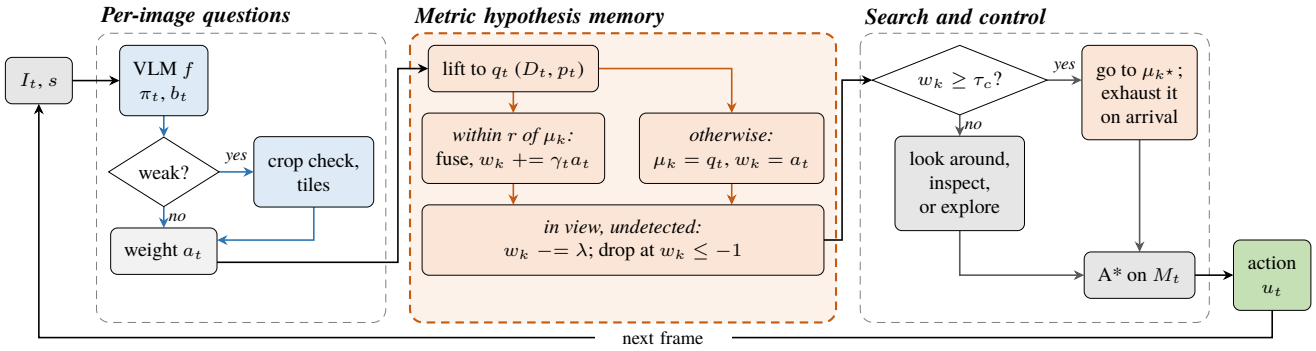


Figure 3. One step of S2M. The frozen VLM f gives the presence π_t and box b_t of the phrase s on the frame I_t; a weak answer is checked on a crop or on tiles, which sets its weight a_t. The box is lifted to a floor point q_t with depth D_t and pose p_t and fused into a hypothesis μ_k within r, or starts one; $\gamma_t < 1$ discounts a repeated view, and a hypothesis that should be seen but is not loses λ of its evidence w_k. Above τ_c a hypothesis is committed, and the agent goes to $\mu_{k^\star}$; A* on the map M_t gives the action u_t.

4. Semantic-to-Metric Navigation

S2M is a navigation agent that instantiates the factorization of Eq. (3): the VLM is its semantic sensor, and the agent owns all state and every decision. The belief $\mathcal{B}_t = (\mathcal{H}_t, M_t)$ consists of a list of target hypotheses $\mathcal{H}_t$ and an occupancy map M_t; Fig. 3 shows one step in full.

4.1. Per-image questions

At the start of an episode the VLM f rewrites ℓ into a target phrase s: the category when the instruction names or implies one, and the appearance words when it describes only how the object looks. At every step two queries about the current frame run in parallel. The presence probability is read from the first answer token,

$$\pi_t = \frac{P_f(\mathit{yes} \mid I_t, s)}{P_f(\mathit{yes} \mid I_t, s) + P_f(\mathit{no} \mid I_t, s)}, \tag{4}$$

and the second query returns a box b_t for s or none. Neither query contains the action history, the map, or earlier answers, so every answer can be checked against the image alone.

Lifting. A box with centre $c(b_t) = (u, v)$ is lifted with the median depth $\tilde{d}_t$ over its central half (half its width and half its height). With focal length φ and principal point (u_0, v_0) of the pinhole camera, tilt ϕ_t, and the forward and right unit vectors $e_f(\theta_t), e_r(\theta_t)$ of the heading,

$$q_t = x_t + \tilde{d}_t \Big[\big(\cos\phi_t - \tfrac{v-v_0}{\varphi} \sin\phi_t \big)\, e_f(\theta_t) + \tfrac{u-u_0}{\varphi}\, e_r(\theta_t) \Big], \tag{5}$$

which we write $q_t = \Pi_{p_t}^{-1}(c(b_t), \tilde{d}_t)$; lifts with $\tilde{d}_t > 6\,\mathrm{m}$ are discarded. The same back-projection, with the known camera height, labels each depth pixel as floor (height below 0.10 m) or obstacle (0.10–1.9 m) in a 5 cm occupancy grid M_t.

Observation weight. A lifted box enters the memory with weight

$$a_t = \begin{cases} \pi_t & \text{if the target is named and } \pi_t \ge \tfrac{1}{2}, \\ \pi_t^z & \text{otherwise, if } \pi_t^z \ge \tfrac{1}{2}, \\ 0.2\,\pi_t^z & \text{otherwise,} \end{cases} \tag{6}$$

where π_t^z is the presence probability of a *crop check*, a square crop of 2.5 times the box's longer side resized to 448 px. The crop is queried only when presence and box disagree or when the goal is appearance-only, so the extra cost is paid where the evidence is weak.

4.2. Metric hypothesis memory

Each hypothesis k stores a floor position μ_k, evidence w_k, and the poses V_k from which it was seen. An observation (q_t, a_t) is associated with the first hypothesis within $r = 0.5\,\mathrm{m}$. If there is one, then

$$\mu_k \leftarrow \frac{n_k \mu_k + q_t}{n_k + 1}, \qquad w_k \leftarrow w_k + \gamma_t\, a_t, \tag{7}$$

with n_k the number of fused observations and $\gamma_t = 0.25$ if some $v \in V_k$ has the heading of p_t and a position within 0.2 m of it, else $\gamma_t = 1$; otherwise a new hypothesis $(\mu, w) = (q_t, a_t)$ is created. Through γ_t, evidence grows with the number of distinct viewpoints rather than with the number of frames.

Negative evidence. A hypothesis that the current frame should show but that receives no detection loses evidence:

$$\begin{gathered} w_k \leftarrow w_k - \lambda \mathbb{1}\big[\mathrm{vis}_t(\mu_k)\big], \\ \mathrm{vis}_t(\mu) \Leftrightarrow \bar{u} \in \Omega,\ \ d \le 6\,\mathrm{m},\ \ D_t(\bar{u}) + 0.3 \ge d, \end{gathered} \tag{8}$$

for every hypothesis without a detection within r, where $(\bar{u}, d) = \Pi_{p_t}(\mu)$ are its pixel and planar depth, Ω is the image shrunk by 20 px, and $\lambda = 0.4$; the last condition rules

out occlusion. Hypotheses with $w_k \le -1$ are removed; absence outside the view is never counted.

Commitment and exhaustion. The agent commits to a hypothesis once $w_k \ge \tau_c = 0.8$ and keeps it while $w_k \ge \tau_k = 0.4$; the hysteresis prevents one missed detection from abandoning a well-supported target. When the agent stands within 0.9 m of a committed hypothesis and the episode continues, which on EB-Navigation means that it is not the target, the hypothesis is marked exhausted. It stays in memory, so later detections of the same look-alike merge into it instead of creating a new target.

4.3. Search and control

Selection. Among the committed hypotheses $\mathcal{C}_t$ the agent visits

$$k^\star = \arg\max_{k \in \mathcal{C}_t} \frac{1 - e^{-w_k}}{c_k + 1}, \qquad c_k = \frac{\|\mu_k - x_t\|}{0.25\,\mathrm{m}}. \tag{9}$$

The numerator maps the evidence to the probability of at least one detection under a Poisson model with mean w_k, and the denominator counts the steps to reach and check μ_k. For locations with fixed inspection costs, inspecting in decreasing order of this probability-to-cost ratio is optimal [54]; S2M applies the rule greedily, recomputing the cost from the current position. It reduces to nearest-first for equal evidence and to strongest-first for equal cost. With nothing committed, the agent first completes a 360° look-around, then approaches the strongest hypothesis with $w_k \ge 0.35$, and otherwise moves to the observed free cell 0.75–3 m away that is farthest from all visited positions.

Planning. Actions come from A* over (position, heading) on M_t, whose edges are the benchmark's translations and turns. An edge costs 1 plus 0.15 for each of four sampled points on it that is unobserved; edges through inflated obstacles are removed, and a move that the simulator has rejected becomes a forbidden edge.

Magnified search. Small or distant targets are often missed in the full frame. When a view yields no observation with $a_t \ge 0.5$ and nothing is committed, the frame is split into 3×3 overlapping tiles, each upsampled to 448 px and queried for a box in parallel; up to four distinct tile boxes are crop-checked, and those with $\pi^z \ge 0.5$ enter the memory. This happens once per view and spends VLM calls rather than environment steps.

4.4. RGB-only and explicit-stop variants

Monocular metric depth networks err mostly by a frame-dependent scale. Let $\hat{D}_t$ be the raw prediction and $\kappa(v) = \sin\phi_t + \frac{v - v_0}{\varphi}\cos\phi_t$ the downward component of the ray through row v per unit of planar depth. A pixel on the floor satisfies $\sigma\,\hat{D}_t(u,v)\,\kappa(v) = h$ for the camera height h, and an object standing on the floor gives a larger σ. Over the pixels $\mathcal{F}_t$ whose rays point at least 15° below the horizontal, S2M takes the mode

$$\sigma_t = \operatorname*{mode}_{(u,v)\in\mathcal{F}_t} \frac{h}{\hat{D}_t(u,v)\,\kappa(v)}, \qquad D_t = \sigma_t\,\hat{D}_t, \tag{10}$$

with the mode taken over values in $[\bar{\sigma}/1.33,\ 1.33\,\bar{\sigma}]$, where $\bar{\sigma}$ is one global scale fitted on development frames; it keeps $\bar{\sigma}$ when fewer than 10% of the pixels in $\mathcal{F}_t$ fall in this window. This is the ground-plane constraint of monocular visual odometry [64] applied per frame. Pose comes from dead reckoning over the executed actions,

$$\hat{x}_{t+1} = \hat{x}_t + 0.25\,o_t\big(\alpha_f e_f(\hat{\theta}_t) + \alpha_r e_r(\hat{\theta}_t)\big), \tag{11}$$

with $(\alpha_f, \alpha_r) \in \{(\pm 1, 0), (0, \pm 1)\}$ the translation of u_t, and heading and tilt advanced by the turn or tilt of u_t when $o_t = 1$.

Explicit stopping. EB-Navigation ends an episode on success; RoboTHOR requires a Stop action. There the agent stops only when it is within 0.9 m of a committed hypothesis, has turned within 20° of it and tilted it into view, and a fresh presence query exceeds 0.5; otherwise the hypothesis is exhausted. Because RoboTHOR episodes last up to 500 steps, negative evidence counts only within 3 m and once per view, and exploration goes to the nearest frontier.

Information access. The VLM sees the current image, a crop or tile of it, and s; the controller sees depth, pose, and action success.

5. Experiments

Setup. We evaluate on the full EB-Navigation test set (300 episodes, official environment, actions, and success test) and on the full RoboTHOR ObjectNav validation split [13] (1800 episodes, 15 apartments, 12 categories; stochastic LoCoBot actuation, 30° turns, 500 steps, and a Stop that succeeds within 1 m of a visible target-category object). The models are Qwen3-VL-4B-Instruct [70] and Qwen2.5-VL-3B-Instruct, served with vLLM [71] at temperature 0 on one 8 GB RTX 5060 that also renders the simulator; S2M uses their 4-bit AWQ checkpoints. The EB prompt uses the benchmark's prompts, three examples, parser, and plan execution. Monocular depth comes from Depth Anything V2 Metric-Indoor-Base [59] on the CPU. *Protocol.* S2M was run four times on the test set and the RGB-only configuration twice, each entering with its per-episode mean. All comparisons are paired over the 300 episodes, with bootstrap 95% CIs and sign-flip permutation tests, Holm-corrected over the ten variants of Sec. 5.2.

5.1. Main results

With Qwen3-VL-4B, S2M succeeds in 79.4% of the episodes against 55.0% for the EB prompt (Tab. 1), a paired gain of +24.4 pp (CI [+18.0, +30.8], $p<10^{-5}$). The gain

Table 1. Success rate (%) on EB-Navigation test. Top: reported results. Bottom: our runs on one 8 GB GPU, with S2M at 4-bit. ±: std. over test runs. Bold: best per model.

Method	Model	Base	Common	Complex	Visual	Long	Avg.	95% CI
Prompted MLLMs (EmbodiedBench)								
EB prompt [1]	GPT-4o	55.0	60.0	58.3	60.0	55.0	57.7	–
EB prompt [1]	InternVL3-78B	66.7	63.3	61.7	45.0	31.7	53.7	–
EB prompt [1]	Gemini-2.0-flash	63.3	65.0	50.0	51.7	13.3	48.7	–
EB prompt [1]	Claude-3.7-Sonnet	50.0	61.7	50.0	36.7	26.7	45.0	–
EB prompt [1]	Qwen2.5-VL-72B	46.7	46.7	46.7	26.7	33.3	40.0	–
EB prompt [1]	InternVL3-8B	38.3	30.0	40.0	33.3	5.0	29.3	–
EB prompt [1]	Qwen2.5-VL-7B	20.0	26.7	38.3	16.7	0.0	20.3	–
Training-free agents								
MOSAIC (module graph) [67]	Qwen3-VL-8B	60.0	46.7	45.6	51.1	46.7	50.0	–
ESCA (scene graph) [5]	GPT-4o	66.7	62.0	63.3	55.0	26.7	54.7	–
EmbodiSkill (skills) [6]	Qwen3-VL-32B	65.0	68.0	73.0	67.0	33.0	61.3	–
HODAgent (agent loop) [68]	Qwen3.6-35B-A3B	76.7	55.0	68.3	55.0	36.7	58.3	–
RoboFoundry (self-evolving) [69]	Qwen3.8-27B	76.7	85.0	73.3	63.3	43.3	68.3	–
BrainMem (memory) [3]	GPT-4o	73.3	71.7	80.0	66.6	63.3	71.0	–
EMHO (harness) [2]	Qwen3.8-27B	86.7	76.7	73.3	76.7	63.3	75.3	–
RoboFoundry (self-evolving) [69]	GPT-6 Astra	90.0	85.0	86.7	73.3	66.7	80.3	–
Tool library [4]	Qwen3-VL-8B	85.0	85.0	77.0	75.0	77.0	80.0	–
Tool library [4]	GPT-4o	80.0	85.0	85.0	80.0	83.0	83.0	–
Fine-tuned								
SVA (Q-model) [7]	Qwen3.5-4B	54.2	41.7	45.8	41.7	50.0	46.7	–
MuSix (world models) [8]	Qwen3-VL-4B	63.2	63.2	57.6	62.4	43.2	57.9	–
Ours (one 8 GB GPU)								
EB prompt	Qwen3-VL-4B	66.7	65.0	65.0	**53.3**	25.0	55.0	[49.3, 60.7]
S2M	Qwen3-VL-4B	**89.6**	**84.2**	**88.3**	52.9	**82.1**	**79.4** ±0.5	[75.0, 83.7]
S2M, RGB only†	Qwen3-VL-4B	85.0	80.8	85.8	52.5	78.3	76.5 ±0.2	[71.8, 81.0]
EB prompt	Qwen2.5-VL-3B	21.7	21.7	28.3	20.0	10.0	20.3	[16.0, 25.0]
S2M	Qwen2.5-VL-3B	**78.3**	**71.7**	**71.7**	**53.3**	**75.0**	**70.0**	[64.7, 75.0]
Oracle target‡	–	93.3	93.3	93.3	93.3	83.3	91.3	[88.0, 94.3]

†No depth sensor and no simulator pose (Sec. 5.3). ‡Geometric planner given the true target position: an upper bound.

follows the predictions of Sec. 3. It is largest where the target starts behind the agent (ii): on Long, S2M succeeds in 82.1% against 25.0% (Fig. 5a), 1.3 pp below the oracle, and it is +22.9, +19.2, and +23.3 pp on Base, Common, and Complex. On Visual, the smallest gain with both models, the two agents are level (iii; −0.4 pp, CI $[-15.0, +14.6]$, p=1.00; Sec. 5.4). With Qwen2.5-VL-3B, S2M reaches 70.0% against 20.3% (+49.7 pp, CI [+43.0, +56.3]): the smaller model costs S2M 9.4 pp and the EB prompt 34.7 pp (iv).

Efficiency. On the 140 episodes that both agents solve in every run, S2M needs 11.1 steps against 12.6 (p=1.9×10^{-4}); within ten steps it solves 28.2% of the episodes against 16.0% (Fig. 4a).

Comparison with published agents. With a 4B model and no training, S2M exceeds every prompted MLLM up to GPT-4o and the scene-graph, module-graph, skill, memory, harness, and self-evolving agents built on GPT-4o and 8–35B models [2, 3, 5, 6, 67–69], and MuSix and SVA, which train extra models around a 4B policy [7, 8]. Only the tool-library agents [4] and RoboFoundry on GPT-6 Astra score higher (Fig. 6).

Where the gain comes from. The diagnosed failure modes largely disappear (i). Per run, S2M never sees the target in 0.8 episodes (EB prompt: 12), is blocked in 0.5 (26), loses a seen target in 28.5 (43), and ends near it in 9.8 (34) (Fig. 4b, Fig. 5): the look-around, the occupancy map, and the memory of the target remove these failure modes.

5.2. Components

Table 2 changes one element at a time. Disabling multi-view fusion, so that each detection replaces the earlier hypotheses, lowers success by 10.4 pp (CI $[-15.0, -6.0]$, Holm p=8.1×10^{-5}), the only component whose removal survives the correction; the loss falls on the four subsets in which the agent starts facing the target. Fusion does not help the agent

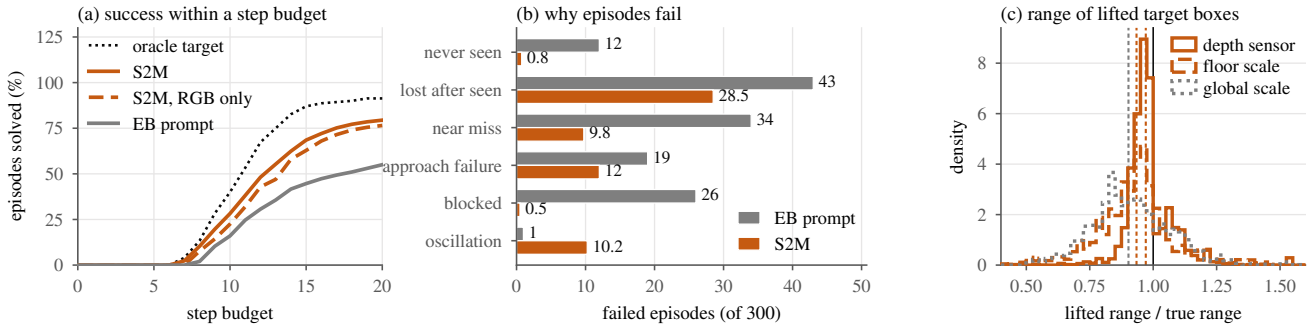


Figure 4. EB-Navigation test, Qwen3-VL-4B. **(a)** Episodes solved within a step budget, averaged over runs. **(b)** Failed episodes by outcome (first matching rule, Sec. 3; mean over the runs of S2M). **(c)** Lifted range divided by true range for detections within 1 m of the target, with medians (dashed), for S2M with the depth sensor and with monocular depth under a global and a per-frame floor scale (Eq. (10)).

Table 2. Components and sensing on EB-Navigation test (Qwen3-VL-4B). Δ: paired difference from the full model; p: sign-flip permutation test, Holm-corrected. Bold: significant.

Variant	Base	Common	Complex	Visual	Long	Avg.	Δ (95% CI)		p
S2M (full)	89.6	84.2	88.3	52.9	82.1	79.4	–		–
Components									
w/o fusion	78.3	70.0	75.0	41.7	80.0	69.0	**−10.4**	[−15.0, −6.0]	8.1×10^{-5}
w/o look-around	95.0	91.7	91.7	71.7	48.3	79.7	+0.2	[−4.2, +4.8]	1.00
w/o exhaustion	88.3	85.0	88.3	46.7	83.3	78.3	−1.1	[−2.5, +0.2]	0.89
w/o neg. evidence	93.3	83.3	88.3	58.3	81.7	81.0	+1.6	[−1.2, +4.5]	1.00
w/o tiled search	86.7	80.0	85.0	60.0	80.0	78.3	−1.1	[−4.5, +2.4]	1.00
w/o crop check	85.0	80.0	83.3	51.7	83.3	76.7	−2.8	[−5.7, +0.1]	0.57
strongest first	90.0	85.0	90.0	55.0	83.3	80.7	+1.2	[−0.3, +2.9]	0.89
Sensing									
odometry noise (5%, 2°)	91.7	81.7	90.0	50.0	83.3	79.3	−0.1	[−3.1, +2.9]	1.00
RGB only, global scale	56.7	58.3	60.0	36.7	53.3	53.0	**−26.4**	[−32.4, −20.5]	$<10^{-5}$
RGB only, floor scale	85.0	80.8	85.8	52.5	78.3	76.5	−2.9	[−6.3, +0.3]	0.67

find the target; it keeps the target once found: 74% of single lifted detections lie within 0.5 m of it, and a hypothesis there reaches commitment in 264 episodes without fusion and 258 with it. Without fusion, each detection, including the false boxes of Sec. 3, replaces the committed hypothesis, and lost targets rise from 28.5 to 42 episodes and near misses from 9.8 to 20. Removing the look-around leaves the average unchanged (0.2 pp) but trades one subset against the others: Long changes by −33.7 pp because the agent no longer turns to find a target behind it, while the other subsets gain the steps that the turn no longer costs (+3.3 to +18.8 pp). Removing exhaustion on arrival changes Visual by −6.2 pp, where appearance-only goals produce look-alikes that must be ruled out once visited. Negative evidence, tiled search, the crop check, and the search index each change success by less than 3 pp, none significantly. They cost no environment steps and matter when candidates compete, which EB-Navigation's single rooms rarely produce.

5.3. Sensing

Pose. Dead reckoning (Eq. (11)) is exact in heading and tilt and errs by 0.10 m in position on average, from the simulator's grid snapping. Accumulated odometry noise of 5% per translation and 2° per turn changes success by −0.1 pp (Tab. 2).

Depth. Monocular depth with one global scale $\bar{\sigma}$ is not enough: with pose from the executed actions, S2M reaches 53.0%, level with the EB prompt (−2.0 pp, p=0.65). The global scale places target detections at a median 0.90 of their true range (Fig. 4c) and lifts 42.5% of the nearby floor above the obstacle threshold, so the agent walks to a point short of the target: 66 failures exhaust a hypothesis within 1 m of it, none with the sensor. The per-frame floor scale (Eq. (10)) lowers the falsely blocked floor to 9.9% and moves the range ratio to 0.93 (sensor: 0.97); success rises to 76.5%, +21.5 pp over the EB prompt ($p<10^{-5}$) and 2.9 pp below the depth sensor (Holm p=0.67).

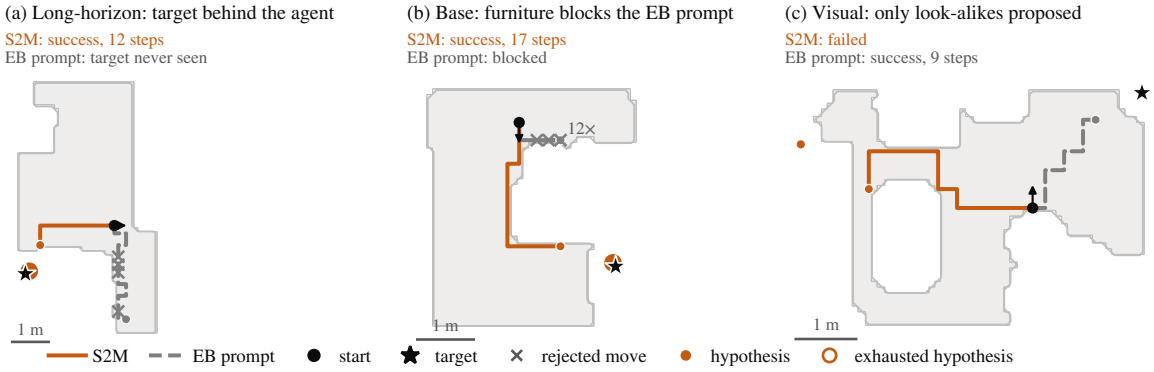


Figure 5. Trajectories on three EB-Navigation test episodes (Qwen3-VL-4B), with the hypotheses of S2M at the end of the episode sized by evidence. Grey: reachable floor.

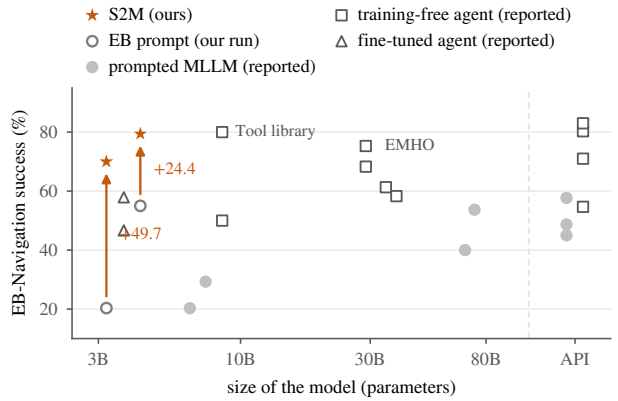


Figure 6. EB-Navigation success against model size (Tab. 1). Arrows: gain of S2M over the EB prompt.

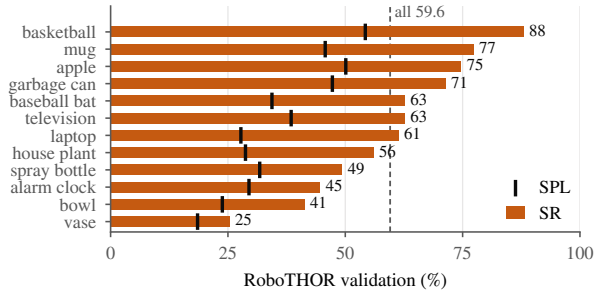


Figure 7. S2M on RoboTHOR validation by target category.

5.4. Remaining gap: appearance-only goals

Visual accounts for 68% of the gap to the oracle and is the only subset on which S2M does not exceed the EB prompt. The failure is recognition: a candidate comes within 0.5 m of the target in 93–98% of the episodes of every other subset but in 63–67% of the Visual ones, and 65 of the 113 Visual failures never produce one.

Table 3. RoboTHOR ObjectNav validation (1800 episodes). Bold: best; underlined: second. §From SG-Nav [11].

Method	Input	Year	SR	SPL
Zero-shot				
CoW [9]	RGB-D	2022	26.7	16.9
LGX [72]	RGB-D	2024	35.0	21.9
ESC [73]	RGB-D	2023	38.1	22.2
L3MVN§ [74]	RGB-D	2023	41.2	22.5
VLFM§ [10]	RGB-D	2023	42.3	23.0
OpenFMNav§ [75]	RGB-D	2024	44.1	23.3
SG-Nav [11]	RGB-D	2024	47.5	24.0
UniGoal [28]	RGB-D	2025	48.0	24.2
CogNav [30]	RGB-D	2024	54.6	24.3
MVP-Nav [76]	RGB	2026	57.5	26.8
ForesightNav [77]	RGB-D	2026	62.5	32.9
STRIVE [31]	RGB-D	2025	**68.1**	**36.3**
Trained for the task				
ProcTHOR [9, 78]	RGB	2022	<u>66.4</u>	27.4
S2M (Qwen3-VL-4B)	RGB-D	–	59.6	<u>35.9</u>

5.5. RoboTHOR ObjectNav

On the full RoboTHOR validation split, S2M reaches 59.6% success and 35.9% SPL. Among zero-shot methods (Tab. 3) its SR is below only ForesightNav (12B [77]) and STRIVE (Gemini-2.0-flash [31]), and its SPL only below STRIVE. Figure 7 gives each category.

6. Conclusion

Small VLMs fail as navigators because agents make them keep spatial state in text, not because they cannot see. S2M makes the model a semantic sensor and keeps the state in metric form; on one 8 GB GPU, a training-free 4B model matches agents built on far larger ones. Appearance-only goals (Sec. 5.4), whole houses, and real robots remain open.

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